



Saveetha Institute of Medical and Technical Sciences
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Course Code: CSA1711

Slot: C

Course Name: Artificial Intelligence

Course Faculty: Dr. Krishna Kumar

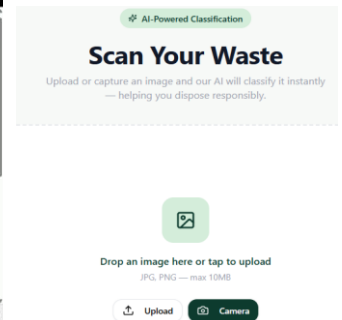
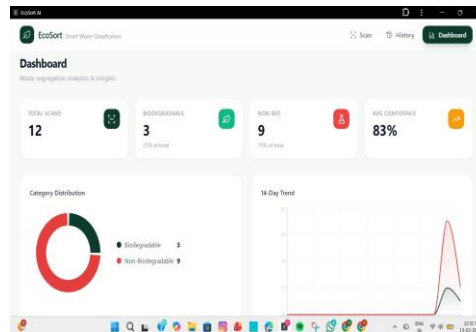
Project Title: Design and Development of a Smart Waste Management System with Automated Trash Segregation

Module Photographs:

Review



Output



Project Description:

Module – Waste Image Collection & Preprocessing

The Smart Waste Classification System is an AI-based system developed to automatically identify and classify waste into different categories such as biodegradable and non-biodegradable waste. The system uses image processing and a Convolutional Neural Network (CNN) to analyze waste images and predict their category. It helps reduce manual waste segregation, improves recycling efficiency, supports proper waste disposal, and contributes to a cleaner and more sustainable environment.

The Waste Image Collection & Preprocessing module is the initial stage of the system, where images of different waste materials are collected from datasets, cameras, or user uploads. The collected images are organized according to their waste categories and processed to remove unwanted or unclear images. The images are resized into a suitable format, normalized, and enhanced using techniques such as rotation, flipping, and zooming. These preprocessing steps improve image quality and prepare a reliable dataset for training and testing the CNN-based waste classification model.

Student Signature

Guide Signature